

Stress loading and matrix remodeling in tendon and skeletal muscle: Cellular mechano-stimulation and tissue remodeling

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Introduction—the concept of mechanical loading on tissue and cells

The loading of connective tissue is a dominant part of all body movement, from light recreational walking to high-level running and athletic activities, such as jumping or throwing. The connective tissue has to resist high levels of loading from muscular contractile activity and, clearly, very high loads will result in acute injuries of connective tissue structures like ligaments, tendons or bones. These present themselves as a tissue rupture or fracture due to a single mechanical load that surpasses the load tolerance.

With regards to repeated loading on the connective tissue there is a fine, yet undefined, line between tissue tolerance of loading and potential adaptation and the development of tissue pathology and associated clinical symptoms. It is, however, clear that very intense physical training, in elite level sports requiring repeated movements of specific body parts, does demonstrate a kind of upper limit of tolerance in the body. Table 4.1 illustrates how elite athletes within

sports, including running, swimming and rowing, experience loading of the lower extremity, shoulder or upper body of up to 40 000–50 000 repetitions per week. Clearly, genetic differences will provide a varying degree of tolerance towards repetitive loading but, in general, it is accepted among athletes that if you extend the amount of training further than that given in Table 4.1, most athletes will experience decreased performance within their sports. In general, this decrease in performance is unlikely to be a result of the total number of training hours, as athletes in several sports that require a more varied loading of the body than those listed in Table 4.1, for example, triathletes, are able to train for up to 35–38 hours per week without any resulting decrease in performance or accumulation of injuries. Thus, Table 4.1 seems to show that each specific regional connective tissue has an upper limit for repeated loading.

Connective tissue in the body is able to withstand substantial loads, varying from clear tensile loading to more compressive loading. Data, primarily from animal literature but also more

Table 4.1 Tissue tolerance in elite athletes

	Training-dose/week	Tissue loading reps/week
Distance running	120 km/week (steps 1.5 m)	~40 000 steps/leg
Swimming	3–4 hours/day (30–40 strokes/min)	~26 000 strokes/arm
Rowing	3 hours/day (22–35 strokes/min)	~38 000 strokes

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recently from human experiments, shows that increased loading results in improved size and strength of tendons, ligaments, bone and cartilage (Table 4.2). It is, however, clear that the improvement is very moderate and that the adaptation requires considerable time. Compared with skeletal muscle adaptation, the response to training in matrix tissue is moderate. On the other hand, findings regarding the influence of inactivity upon different connective tissues normally indicate some kind of loss in tissue strength and mechanical characteristics (Table 4.2). These findings illustrate that connective tissue is highly dependent upon normal daily loading and is very vulnerable when not being mechanically loaded. When subjected to inactivity, the connective tissue does not diminish quickly in size, but does demonstrate a quick (within 1–2 weeks) alteration in passive mechanical properties. The explanation for this is currently unknown but indicates that molecular structures, which are of importance in regard to mechanical properties, can change quickly when unloaded. In artificial tendons it has been shown that just a few days of unloading in tendon-like structures results in a disorganization of the fibrils. In humans,

the quick changes in the mechanical properties of tendon are accompanied by a change in the expression of enzymes of importance for the formation of cross-link molecules.

Mechanotransduction—signaling and outcome

The mechanical loading of connective tissue is a complex process that involves several steps, from the initial conversion of mechanical stress upon the tissue into chemical signals. Initially, integrin receptors will be activated, and this will result in adhesion and activation of mechanotransduction pathways (Kjaer, 2004; Magnusson et al., 2010). An important pathway for mechanotransduction is the Rho-Rock pathway, and the importance of this pathway for mechanotransduction has been documented in several studies. The resulting activation of protein synthesis from the cell nucleus leads to the formation of matrix proteins, such as collagens, and subsequent tissue structure changes, either more dense tissue, larger tissue volume, or altered organization of the fibrillar and other structures, ultimately contributing to the mechanical properties of the matrix tissue (Figure 4.1).

Table 4.2 Examples of connective tissue adaptation to long-term physical training and inactivity				
Function	Capacity (acute)	Training (% increase)	Inactivity (% decrease 4–8 weeks)	
Connective tissue		Months		
Tendon	100 MPa	10	30	(+ stiffness/cross-links)
Ligament	60–100 MPa	10	30	(+ stiffness/elastin)
Bone	50–200 MPa	5–10	30	(+ mineralization/Ca++)
Cartilage	5–40 MPa	5–10	30	(+ water/proteogl)
Skeletal muscle		Months		
Strength		100	60	
Muscle mass		60	20–30	

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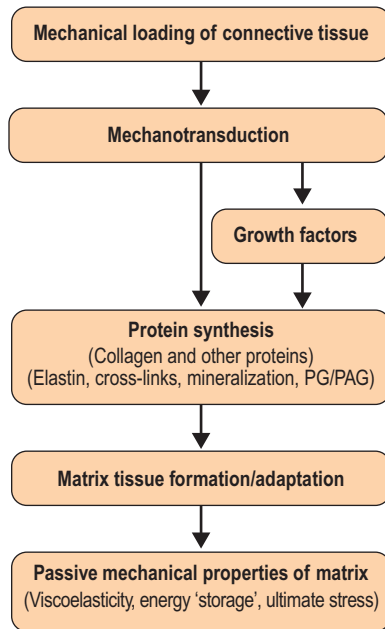


Figure 4.1 Mechanical loading of connective tissue

Mechanical loading will also lead to local upregulation (i.e. more pronounced gene expression and increased synthesis of the protein) of growth factors, such as IGF-I and TGF- β . These will probably be released from the fibroblasts of the connective tissue. Whether these growth factors will act in an autocrine or a paracrine fashion has yet to be explained, but the upregulation of these factors was found to be associated with an exercise-induced upregulation of collagen protein synthesis. This suggests a direct association between growth factor release and matrix protein formation (Figure 4.1). There is also some indication that mechanical loading and growth factors can stimulate the matrix tissue in an additive or even synergistic way (Magnusson et al., 2010).

Mechanical loading increases both collagen expression and protein synthesis in animal

studies (Heinemeier et al., 2007). In humans, an upregulation of protein synthesis has been demonstrated both in peri-tendinous tissue and, determined from the incorporation of labeled amino acids, in tendon tissue (Miller et al., 2005). Furthermore, a degradation of collagen tissue has also been demonstrated in humans. Despite these rather dynamic changes of collagen turnover in tendon, this does not prove any major exchange of tendon structure with mechanical loading. Very recent data indicate that the exchange of tendon structures takes place in the first 17 years of life and that, thereafter, a more stable structure is maintained in an intact uninjured tendon (Heinemeier et al., 2013). The fact that well-trained individuals have higher cross-sectional areas of Achilles and patella tendons than untrained ones, and that within the same individual one stronger leg also shows a larger diameter tendon than on the contralateral side (Couppe et al., 2008), indicates that either the growth of tendon with training occurs in the early years, or that it represents an addition of tissue on the surface of the tendon, not unlike the addition of layers on a tree from year to year.

Cellular responses to mechanical loading: *in vitro* to *in vivo*

During the mechanical loading of connective tissue, for example, tendons, the cell within the tendon responds, as previously described, with an increase in mRNA and protein synthesis in the central elements of the matrix (collagen type I). The studies show that interstitial concentrations of collagen propeptide fragments in the peritendinous space rose after exercise. Furthermore, when infusing amino acids labeled with stable isotopes, and determining their incorporation into the relevant connective tissue, it has been shown that this incorporation into hydroxyl-proline, and thus into collagen,

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almost doubled in both tendon and skeletal muscle after intense exercise (Miller et al., 2005). Conversely, with inactivity over 2–3 weeks, it can be shown that such incorporation would be diminished. Therefore, it appears that collagen production in tendon and skeletal muscle matrix is affected by the degree of mechanical loading. Interestingly, both in tendon and muscle, it seems that the dose-response curve for the relationship between the intensity of loading and the responding protein synthesis outcome levels off relatively early. This means that, already at a relatively moderate loading of the tissue, a sufficient connective tissue response is observed. As adaptation of connective tissue in general is relatively slow, it represents an advantage that also relatively moderate loading will result in an increase in protein synthesis. This means that during rehabilitation after an injury, humans may stimulate their connective tissue with low levels of intensity, at a time when the tissue would otherwise be too weak to tolerate heavy muscle resistance training.

The question now is whether or not this reflects a true tissue renewal and replacement of existing fibrillar structures, or if it represents a mechanism by which mechanical loading provides collagen for potential incorporation in the case of injury. It could be that only a minor fraction of adult tendon is turned over and that a large part of the tendon is more inert and, therefore, less dynamic. To determine this, an experiment can be performed, in which the atmospheric content of the radioactive carbon isotope ^{14}C can be used to mark the age of connective tissue. This is possible because the content of ^{14}C peaked with atom bomb trials in the late 1950s and early 1960s, after which it was banned and the ^{14}C content in the air dropped in subsequent years. Using this methodology, it can be shown that,

by taking into account the increased turnover of tissue in the first 17 years of life during height growth, it would appear that no major collagen turnover occurred in the adult tendon (Heinemeier et al., 2013). This implies that, despite loading, the tendon remains relatively inert during adult life and that only mechanical damage can cause changes. Therefore, it is suggested that the cells in the adult tendon are predominantly dormant and will only be activated in emergency situations.

When the adult tendon is prepared so that cells are isolated and grown in 3D culture systems where, due to attachment patterns, they will be subjected to tension, it is observed that new and artificial tendon constructs are formed. These artificial tendons are cell-rich, produce new fibrils, display an aligned fibril structure, and have the same qualitative type of mechanical properties as natural cells. They express proteins that are important for tendon structure, such as fibrillar collagens, as well as proteins that are phenotypical for tendons, like tenomodulin. Interestingly, when tension is released from these tendon constructs, they all lose their expression of collagen, tenomodulin and mechanosensitive integrin receptors. These changes do not seem to be compensated for by the excess amount of growth factors added (e.g. TGF- β). They indicate that mechanical loading is crucial for the stimulation of important proteins for matrix structure formation.

Chronic effect of mechanical loading on matrix cells and tissue

To what extent the adult tendon can adapt to physical training is not fully understood. In cross-sectional studies, it has been shown that endurance runners have thicker Achilles tendon

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structures than untrained weight-matched counterparts. Furthermore, when comparing athletes competing in different types of sports, it was been found that runners, who repetitively load their calves, and volleyball players, who use explosive jumping activity, display thicker Achilles tendons than kayak rowers, who load their legs less but have a similar athletic status (Magnusson et al., 2010). In elite older runners, the cross-sectional area of the patella tendon is larger than in untrained age-matched males. The fibril volume was also larger. This indicates that differences between trained and untrained tendons cannot be explained by water accumulation. This data could, therefore, imply that selection is the cause of these differences. The theory of adaptation being related to training is further supported by research into sports with different loading on two legs, such as fencing and badminton. In these examples, where the quadriceps muscle strength was greater on one leg, it was accompanied by a larger patella tendon cross-sectional area (Couppé et al., 2008). To what extent these adaptations of tendon have occurred very early in life or throughout life is not known.

In skeletal muscle, regular training is not associated with any marked increase in collagen content but extreme loading does increase collagen synthesis. This could indicate that collagen degradation increases and that the total collagen content does not, therefore, increase as a result of training. However, this does not indicate to what extent the intramuscular connective tissue will obtain altered mechanical properties with training that is unrelated to collagen content but is, rather, related to fibril arrangement or the synthesis of other molecules related to tissue mechanical properties, for example, cross links (Figure 4.2).

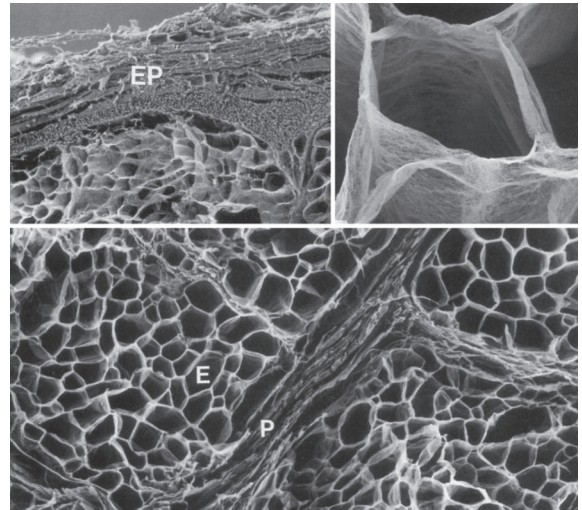


Figure 4.2 Scanning electron micrographs of intramuscular connective tissues (bovine semitendinosus muscle after removal of skeletal muscle protein)

Upper left: epimysium (EP); upper right: endomysium surrounding an individual muscle fiber; bottom picture: perimysium (P) plus endomysium (E).

Modified from Nishimura et al. (1994) with permission.

The borderline between physiological and pathological responses to loading

How much is considered to be adequate loading of connective tissue and how much is too much is an important but, to date, an unresolved question. We do know that appropriate restitution between training bouts will contribute to allow full stimulation of protein synthesis and protein degradation, in order to avoid a gradual net loss of connective tissue over time. Whether this mismatch is the major “road” leading to matrix tissue overload is not completely clear. The presence of both apoptotic cells, and the release of heat shock proteins and inflammatory substances with heavy tissue loading, could indicate that cells in an overloaded matrix tissue are changing their task from matrix protein production, and adequate maintenance

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and renewal of tissue, towards a role where the struggle to survive is their primary goal (Millar, 2005). It may also be that, in the case of tissue overloading, the overloaded cells are shielded from further loading and that this will, therefore, result in further degenerative local changes due to a lack of mechanical loading (Arnoczky et al., 2007). Interestingly, the histological and microstructural changes in early overloading of a tendon, for example, do not mimic the picture of an experimental rupture of a tendon. Therefore, this must represent some other changes behind the overuse of connective tissue. It is known that in some tissues, like tendons, the overuse of the tissue is associated with occurrence of neovascularization and the ingrowth of nerve structures that will result in painful symptoms from the overloaded tissue (Magnusson et al., 2010).

Influence of maturation and aging

Compared with skeletal muscle fibers, our understanding of the effects of aging on related connective tissues is scarce. Although these tissues seem to become more prone to injuries with

aging, the underlying mechanisms have not been clarified. While there is not complete agreement in the literature, the current *in vitro* data indicate that aging tends to accompany a decreased potential for cell proliferation, as well as a reduction in the number of stem/progenitor-like cells. In addition, it seems that turnover in the core of the tendon after maturity is either absent or very small. By contrast, collagen content, whole tendon size and fibril diameter appear to be fairly stable during aging. The most striking feature of aging is an associated severe increase in glycation-derived cross-links.

Major mechanical, cellular and morphological changes already appear to occur during maturation. Aging appears to be associated with a reduction in failure strength and in stiffness (i.e. decreased elastic modulus; see Figure 4.3). These changes in mechanical properties may contribute to increased injury risk, although the details of this influence remain to be elucidated.

A recent magnetic resonance imaging investigation examined the muscular connective tissue

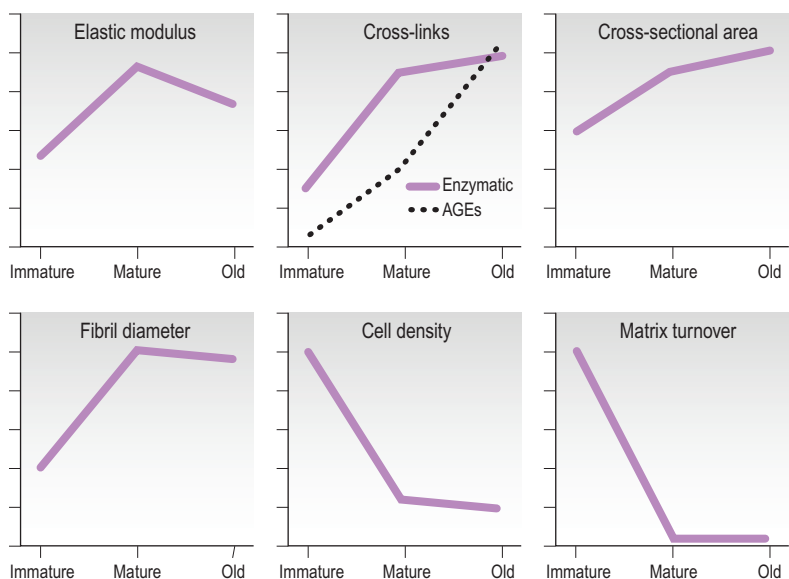


Figure 4.3 Simplified overview of changes observed in tendon tissue with maturation and aging

The graphs are not representations of specific data, but rather indicate the general trends. Highest values of each parameter are oriented towards the top of the graph while the remaining line portions represent the relative trends. AGEs, advanced glycation end products.

Illustration from Svensson et al. (2016) with permission.

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of the thigh in elderly (aged ~65 years) and younger (aged ~25 years) individuals who were either endurance-trained or had a sedentary lifestyle (Mikkelsen et al., 2017). The total cross-sectional area of non-contractile intramuscular tissue (fat and fibrous connective tissue) was found to be greater in older but lower in trained individuals. Subcutaneous adipose tissue was also lower in trained individuals, but was apparently not influenced by aging. The amount of fibrous connective tissues was larger in the two older groups and was lower the two trained groups (Figure 4.4). The mean proportion of these tissues in older, untrained males was 2.3-fold higher than in young, untrained males.

In vastus lateralis biopsies, no influence of age or training was found on levels of endomysial connective tissue, whereas the organization of the perimysium appeared as more complex in the two older groups. No clear difference related to training for intramuscular inflammatory signaling was seen in these samples. Gene expression of IL6 and TNF α was not different among groups, while IL1-receptor and TNF α -receptor 1 levels were lower with age.

It was concluded that the association of increased muscular connective tissues with aging seems not to be accompanied by any severe changes in intramuscular inflammatory signaling, but rather by direct regulatory factors for protein synthesis and proteolysis, which may be involved in the age-related loss of muscle mass in humans.

The finding of a higher total area of muscular connective tissues in the older subjects was seen as an indication that the “usual” age-related loss of gross skeletal muscle underestimates the true loss of the contractile muscle part, and thus of muscle force. In short, as we age, we tend to lose more and more contractile muscle fibers and gain more muscular connective tissues.

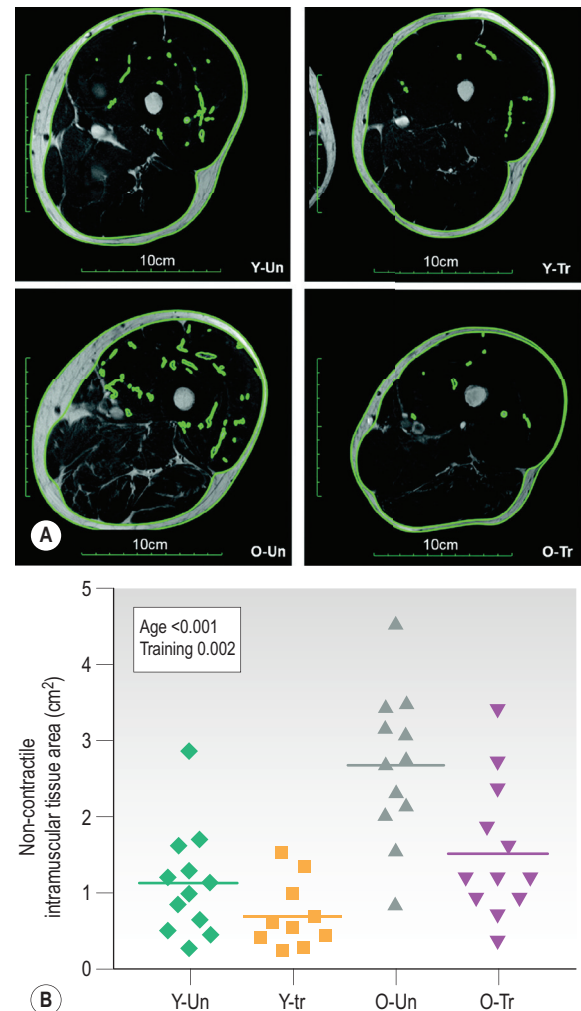


Figure 4.4 Comparison of cross-sectional area of non-contractile connective tissue (NCIT) within and around the thigh musculature (not including the subcutaneous adipose tissue)

(A) Shows representative images from each of the four groups, with NCIT marked in green. (B) Shows individual NCIT with line at mean. Significant effects of training and aging are given for the four groups: endurance-trained runners (O-Tr), older, untrained controls (O-Un), young males matched for current running distance (Y-Tr), and young, untrained controls (Y-Un).

Illustration from Mikkelsen et al. (2017) with permission.

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Clinical summary

The mechanical loading of connective tissue is important for the maintenance and adaptation of the matrix in regards to its composition, structure and passive mechanical properties. The question that remains, however, is what the cells in the connective tissue will sense. Is it strain or is it other types of stress? There is, though, no doubt that connective tissue will respond to mechanical loading, but that the response is very varied, depending upon the magnitude of the load, the type of matrix tissue, and the characteristics of the individual. Clinical work suggests that the overloading of tissue in relation to sports, for

example, represents a challenge in understanding what signs and markers will provide the “warning sign” for overloading. Having said that, it is also clear that, in many clinical situations, the connective tissue is subjected to far too low a mechanical load. From the data present today, it is clear that mechanical loading provides one of the strongest stimuli, if not the strongest, towards an adaptation of matrix tissue that becomes stronger and, in an injury recovery situation, heals faster and better than if no loading were present. The mechanical stimulation associated with regular sports activity may also counteract some of the structural tissue changes of aging.

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